

Batalin–Vilkovisky algebras from Ševera bicomplexes

Domenico Fiorenza

joint work with Eugenia Boffo

Sapienza University of Rome

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The guiding question

Question

Ševera has shown that the canonical BV Laplacian on half-densities of an odd symplectic manifold is naturally obtained as the d_2 differential of a suitable spectral sequence. This is done by explicit local identification with

$$\Delta_{loc} = \sum_i \frac{\partial^2}{\partial p_i \partial q^i}.$$

Can we see that d_2 is a BV operator directly, without relying on the fact we know Δ_{loc} is a BV operator?

The answer has three ingredients:

- 1 Cartan differential operators and their derived brackets;
- 2 spectral sequences with controlled differential-operator order;
- 3 the Ševera bicomplex of an odd symplectic manifold.

Road map

- ① Grothendieck differential operators
- ② Cartan differential operators and BV_0 -modules
- ③ Spectral sequences and Ševera bicomplexes
- ④ The odd symplectic example
- ⑤ The BV algebra of functions and compatible half-densities
- ⑥ Derived Cartan calculus: outlook

Grothendieck differential operators

Let A be a graded commutative algebra and M, N graded A -modules.

$$\mathrm{Diff}_A^{\leq -1}(M, N) = 0,$$

and recursively

$$\mathrm{Diff}_A^{\leq n}(M, N) = \left\{ B \mid [B, \rho_a] \in \mathrm{Diff}_A^{\leq n-1}(M, N) \ \forall a \in A \right\}.$$

Here

$$[B, \rho_a] = B\rho_a - (-1)^{|B||a|} \rho_a B.$$

Key property

The filtration is multiplicative:

$$\mathrm{Diff}_A^{\leq m} \circ \mathrm{Diff}_A^{\leq n} \subseteq \mathrm{Diff}_A^{\leq m+n}.$$

This is the mechanism that will control the order of higher spectral-sequence differentials.

From multiplication to contraction

Put

$$\mathfrak{g}_A = A[-1].$$

The A -module structure on M can be viewed as a contraction

$$\iota : \mathfrak{g}_A \otimes M \longrightarrow M[-1].$$

The Cartan filtration

If M is faithful, identify A with its image in $\text{End}_k(M)$ and define

$$\text{Cartan}_A^{\leq 0}(M, M) = A,$$

$$\text{Cartan}_A^{\leq n}(M, M) = \{B \mid [B, \iota_a] \in \text{Cartan}_A^{\leq n-1} \ \forall a \in \mathfrak{g}_A\}.$$

$$\text{Cartan}_A(M, M) \subseteq \text{Diff}_A(M, M)$$

is a filtered subalgebra.

Derived brackets

For $B \in \text{Cartan}_A^{\leq n}(M, M)$, iterated commutators define

$$[\cdots [[\iota_{a_1}, B], \iota_{a_2}], \dots, \iota_{a_n}] = \iota_{\{a_1, \dots, a_n\}B}.$$

Thus order is measured by the arity of the last nontrivial derived bracket. For order ≤ 2 and

$|B| = 1$:

$$\mathcal{L}_a = [\iota_a, B], \quad [\mathcal{L}_a, \iota_b] = \iota_{\{a, b\}}.$$

If $B^2 = 0$, then the induced bracket on $\mathfrak{g}_A$ is a graded Lie bracket.

BV₀ operators

A degree-one Cartan operator

$$B : M \rightarrow M[1], \quad B^2 = 0,$$

of order at most 2 is a **BV₀-operator**.

It gives a dg Lie algebra morphism

$$\mathcal{L} : (\mathfrak{g}_A, 0, \{-, -\}) \longrightarrow (\text{End}_k(M), [B, -], [-, -]).$$

Moreover, the bracket on A satisfies the shifted Leibniz rule

$$\{a, bc\} = \{a, b\}c + (-1)^{(|a|+1)|b|}b\{a, c\}.$$

Hence A is a P_0 -algebra; after reversing degrees, it is a Gerstenhaber algebra.

Free rank-one modules

Suppose M is free of rank 1 over A , with basis ζ . Then

$$\mathrm{Cartan}_A(M, M) = \mathrm{Diff}_A(M, M).$$

A BV_0 -operator B on M with $B\zeta = 0$ induces

$$\Delta_\zeta : A \rightarrow A[1]$$

through

$$\Delta_\zeta(a) \zeta = B(a\zeta).$$

Then

$$\Delta_\zeta^2 = 0, \quad \Delta_\zeta(1) = 0,$$

and Δ_ζ is second order.

The bracket does not depend on the basis

If $\zeta' = y\zeta$, $y \in A^{0\times}$, the operator Δ_ζ generally changes. But the derived bracket does not:

$$\{a, b\}_{\zeta'} = \{a, b\}_\zeta.$$

Thus

basis-dependent BV operator ; canonical Gerstenhaber bracket.

This is the algebraic version of a familiar BV phenomenon in Physics: the bracket is canonical, while the BV Laplacian requires additional data.

Why spectral sequences?

Let

$$(M, d_0, d_1), \quad |d_0| = (0, 1), \quad |d_1| = (1, 0),$$

be a bicomplex. Suppose (π, K) is a strong deformation retract of (M, d_0) :

$$\mathrm{id}_M - \pi = [d_0, K], \quad \pi K = K\pi = K^2 = 0.$$

The homotopy K supplies canonical primitives in the zig-zag construction of the spectral sequence.

Main idea

Use K to write the higher differentials explicitly on the chain level.

The higher differentials on the chain level

Define

$$\delta_n = \underbrace{d_1 K d_1 K \cdots K d_1}_{n \text{ copies of } d_1}.$$

Its bidegree is

$$|\delta_n| = (n, 1 - n).$$

The n -th spectral-sequence differential is induced by δ_n on the appropriate quotient:

$$Z_n/B_n \cong E_n.$$

The key identities are

$$\delta_k \delta_n = 0$$

and

$$[d_0, \delta_n] = \sum_{k=1}^{n-1} (-1)^{k-1} \delta_{n-k} \pi \delta_k.$$

Controlling differential-operator order

Assume A acts on M and

$$d_0 \in \text{Diff}_A^{\leq 0}, \quad d_1 \in \text{Diff}_A^{\leq 1}, \quad K \in \text{Diff}_A^{\leq 0}.$$

By multiplicativity,

$$\delta_n = d_1 K d_1 \cdots K d_1 \in \text{Diff}_A^{\leq n}(M, M).$$

If Z_n and B_n are A -submodules, the subquotient property gives

$$\boxed{d_n \in \text{Diff}_A^{\leq n}(E_n, E_n).}$$

This is the central abstract result.

Ševera bicomplexes

A Ševera bicomplex consists of

$$(M, d_0, d_1, \pi, K, A, \rho)$$

with

(M, d_0, d_1) a bicomplex,

(π, K) a strong vertical deformation retract,

A a graded commutative algebra acting on M ,

$$d_0 \in \text{Diff}_A^{\leq 0}, \quad d_1 \in \text{Diff}_A^{\leq 1}, \quad K \in \text{Diff}_A^{\leq 0}.$$

We say it has level $\geq N$ if Z_n and B_n are A -submodules for $1 \leq n \leq N$.

Consequence

At level $\geq n$, the spectral-sequence differential d_n has order at most n .

The $n = 2$ payoff

If the Ševera bicomplex has level ≥ 2 and E_2 is free of rank 1 over A , then

$$d_2 : E_2 \rightarrow E_2[1]$$

is a second-order differential operator with

$$d_2^2 = 0.$$

Therefore (E_2, d_2) is a BV_0 -module for A . If ζ is a d_2 -closed basis element, transporting d_2 along

$$E_2 \xrightarrow{\zeta} A$$

gives a BV_0 operator on A .

A useful criterion for level ≥ 2

A sufficient condition is simply

$$d_1 : E_1 \longrightarrow E_1 \quad \text{is zero.}$$

Indeed,

$$Z_2 = \ker d_0, \quad B_2 = \operatorname{im} d_0,$$

so both are A -submodules because d_0 is A -linear. Consequently

$$E_2 = E_1 = H(M, d_0)$$

as A -modules. This is exactly what happens in the odd symplectic example.

Odd symplectic manifolds

Let $(\mathcal{M}, \omega)$ be an odd symplectic manifold of dimension $(n|n)$. Locally there are Darboux coordinates

$$|q^i| = 0, \quad |p_i| = -1,$$

with

$$\omega = \sum_i dq^i \wedge dp_i.$$

The differential forms have a form degree and an internal degree. We regrade by

$$p = f + 2i + n, \quad q = -i,$$

so that

$$d_0 = \omega \wedge - \quad \text{has bidegree } (0, 1), \quad d_1 = d \quad \text{has bidegree } (1, 0).$$

The bicomplex of forms

Because ω has odd total degree,

$$(\omega \wedge -)^2 = 0.$$

Also

$$d^2 = 0, \quad [d, \omega \wedge -] = 0$$

because $d\omega = 0$. Thus

$$(M, d_0, d_1) = (\Omega(\mathcal{M}), \omega \wedge -, d)$$

is a bicomplex. For $A = C^\infty(\mathcal{M})$, multiplication by functions gives the relevant A -module structure.

The Koszul operator Λ

Let Π be the Poisson bivector dual to ω and let

$$\iota_{\Pi} : \Omega \rightarrow \Omega[-2, 1]$$

be contraction by Π . After regrading,

$$\iota_{\Pi} : M \rightarrow M[0, -1].$$

Define

$$\Lambda = [d_0, \iota_{\Pi}].$$

In Darboux coordinates,

$$\Lambda(dq^I \wedge dp_J) = (n - |I| + |J|)dq^I \wedge dp_J.$$

Hence

$$M = \bigoplus_{k \geq 0} V_k, \quad \Lambda|_{V_k} = k \operatorname{id}.$$

The contracting homotopy

The operator d_0 commutes with Λ , and ι_Π preserves the V_k . Define

$$K = \bigoplus_{k>0} \frac{1}{k} \iota_\Pi|_{V_k}, \quad \pi = \text{projection onto } V_0.$$

Then

$$[d_0, K] = \text{id} - \pi.$$

Moreover,

$$\pi K = K\pi = K^2 = 0.$$

Since every V_k is an A -submodule and ι_Π is A -linear,

$$K \in \text{Diff}_A^{\leq 0}.$$

The zero eigenspace

The equation

$$n - |I| + |J| = 0$$

forces

$$|I| = n, \quad |J| = 0.$$

Thus locally

$$V_0 = \{ f(q, p) dq^1 \wedge \cdots \wedge dq^n \}.$$

In particular, d_0 vanishes on V_0 and

$$(V_0, 0) \hookrightarrow (M, d_0)$$

is a quasi-isomorphism. All V_k , $k > 0$, are d_0 -acyclic.

Why d_1 vanishes on E_1

The parity of the regraded first degree satisfies

$$p \equiv k \pmod{2} \quad \text{on } V_k.$$

Hence

$$M^{\text{odd}, \bullet} = \bigoplus_{k \text{ odd}} V_k.$$

But all odd V_k are acyclic. Therefore the odd columns of

$$E_1 = H(M, d_0)$$

vanish. Since d_1 has bidegree $(1, 0)$, it changes column parity, so

$d_1 : E_1 \rightarrow E_1 \text{ is zero.}$

Thus the sufficient criterion gives level ≥ 2 .

The main geometric theorem

For an odd symplectic manifold $(\mathcal{M}, \omega)$, the data

$$(M, d_0, d_1, \pi, K, A, \rho)$$

form a Ševera bicomplex of level at least 2. Consequently,

$$d_2 : E_2 \rightarrow E_2[1]$$

is a second-order differential operator.

The E_2 page: half-densities

A theorem of Ševera identifies

$$E_2 = E_1 = H(M, d_0) \cong \text{Dens}^{1/2}(\mathcal{M}).$$

(locally, a class is represented by

$$f(q, p) dq^1 \wedge \cdots \wedge dq^n,$$

and under changes of Darboux coordinates f transforms as a half-density).

Thus E_2 is a free rank-one A -module, and d_2 gives the canonical BV_0 operator on half-densities.

$$d_2 = \Delta_{\text{Dens}^{1/2}}.$$

Identifying the induced bracket

The abstract construction gives a canonical P_0 bracket on

$$A = C^\infty(\mathcal{M}).$$

We prove that it is the canonical odd Poisson bracket:

$$\boxed{\{f, g\} = \{f, g\}_\Pi = \iota_\Pi(df \wedge dg).$$

The proof uses:

- the explicit representative $\delta_2 = dKd$;
- K being A -linear;
- projection onto V_0 ;
- a Darboux-coordinate calculation.

The Darboux calculation

For $\xi \in V_0$ we may locally take

$$\xi = dq^1 \wedge \cdots \wedge dq^n.$$

Only the dp_i part of dg contributes to $K(dg \wedge \xi)$, and one obtains

$$\pi(df \wedge K(dg \wedge \xi)) = \sum_i \frac{\partial f}{\partial q^i} \frac{\partial g}{\partial p_i} \xi.$$

Adding the exchanged term gives

$$\sum_i \left(\frac{\partial f}{\partial q^i} \frac{\partial g}{\partial p_i} + (-1)^{|f||g|} \frac{\partial g}{\partial q^i} \frac{\partial f}{\partial p_i} \right) \xi,$$

which is exactly $\{f, g\}_{\Pi} \xi$.

From half-densities to functions

A d_2 -closed basis element

$$\zeta \in E_2 \cong \text{Dens}^{1/2}(\mathcal{M})$$

is a nowhere-vanishing half-density satisfying

$$\Delta_{\text{Dens}^{1/2}} \zeta = 0.$$

Such a half-density is often called *compatible*. Transporting d_2 through the identification determined by ζ gives

$$\Delta_\zeta : C^\infty(\mathcal{M}) \rightarrow C^\infty(\mathcal{M})[1],$$

with

$$\Delta_\zeta^2 = 0, \quad \Delta_\zeta(1) = 0, \quad \text{ord}(\Delta_\zeta) \leq 2.$$

The resulting BV algebra

We obtain the chain of structures

$$\begin{array}{c} (\mathcal{M}, \omega) \\ \Downarrow \\ (\Omega(\mathcal{M}), \omega \wedge -, d) \\ \Downarrow \\ E_2 \cong \text{Dens}^{1/2}(\mathcal{M}) \\ \Downarrow \\ (C^\infty(\mathcal{M}), \{-, -\}_\Pi, \Delta_\zeta). \end{array}$$

The bracket is canonical; the BV Laplacian depends on the choice of compatible half-density. After reversing the grading, this is the usual Batalin–Vilkovisky / Gerstenhaber picture.

Derived Cartan calculus: the outlook

The recursive definition of differential operators can be written as iterated pullbacks. For a dg-commutative algebra A , replace

$$\mathrm{Hom}_k(-, -) \quad \text{by} \quad \mathbb{H}\mathrm{om}_k(-, -)$$

and strict pullbacks by homotopy pullbacks. This suggests a dg space of derived Cartan operators

$$\mathbb{C}\mathrm{artan}_A^{\leq n}(M, M).$$

Taking cohomology yields a natural map

$$H(\mathbb{C}\mathrm{artan}_A^{\leq n}(M, M)) \longrightarrow \mathrm{Cartan}_{H(A)}^{\leq n}(H(M), H(M)).$$

Thus chain-level Cartan data can induce genuine Cartan operators on cohomology.

What is a derived Cartan operator of order ≤ 2 ?

At chain level, the Cartan condition is imposed only up to coherent homotopy.

A degree-one operator B which is a cocycle has derived Cartan order ≤ 2 if

$$\mathcal{L}_a = [B, \iota_a] + [d, S_a] + S_{da},$$

has derived Cartan order ≤ 1 , i.e.,

$$[\mathcal{L}_a, \iota_b] = \iota_{\{a,b\}} + [d, T_{a,b}] - T_{da,b} - (-1)^{|a|} T_{a,db}.$$

These are the equations in Nest–Tamarkin–Tsygan noncommutative differential calculus. After passing to cohomology, the homotopies disappear and one recovers an honest Cartan operator of order ≤ 2 .

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Thanks!